

From Acquisition to Assurance

MOBI-QC - A scalable, open-source framework for automated quality control in multimodal brain/body imaging. A raw LSL recording in, a publication-ready quality report out, with no expert in the loop.



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Motivation

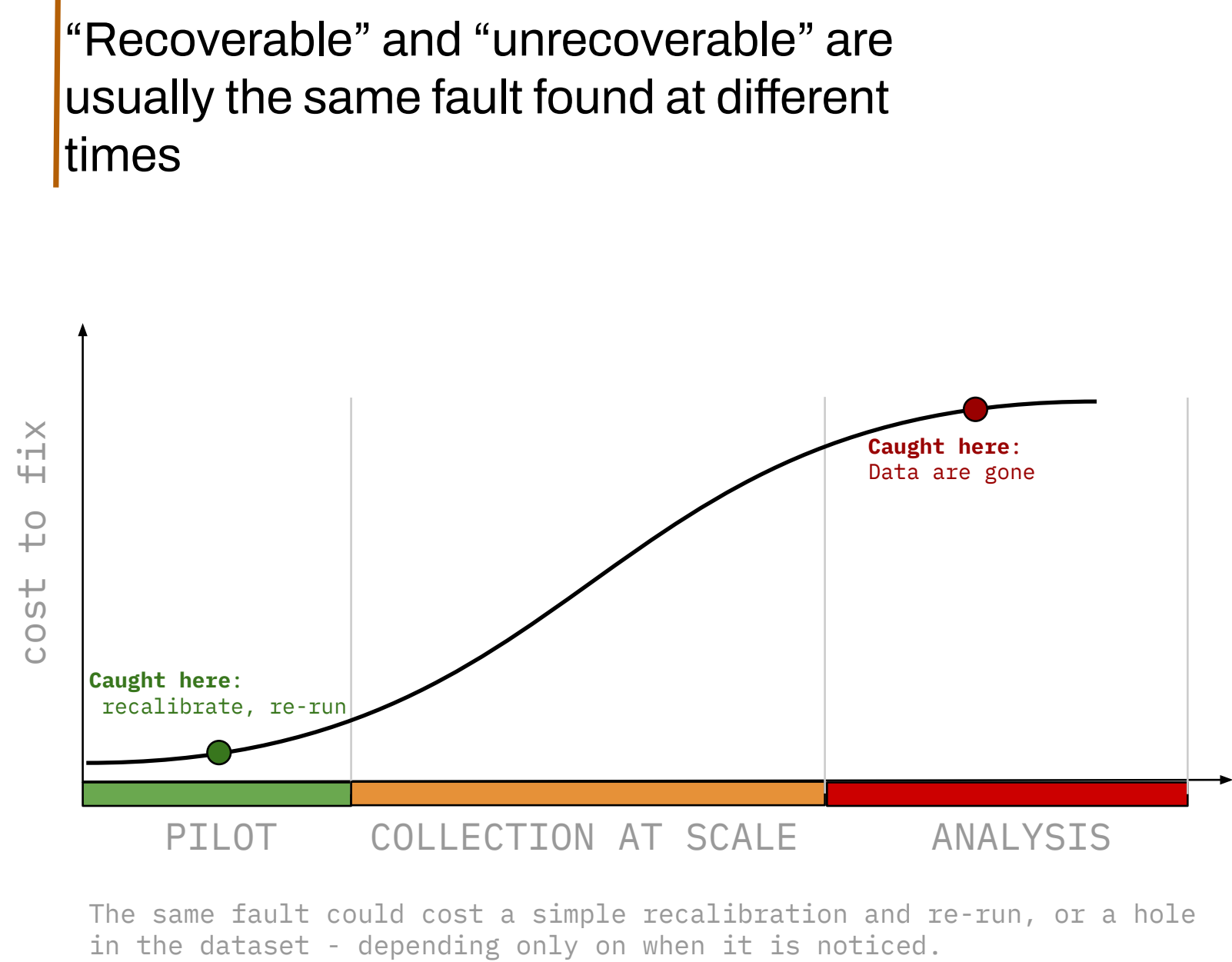
THE PROBLEM

Pilot data looks fine. Then it doesn’t.

Simultaneous collection of heterogeneous streams - distinct hardware systems, software interfaces, and human operators - create numerous opportunities for failures in experimental design, protocol execution and data integrity¹.

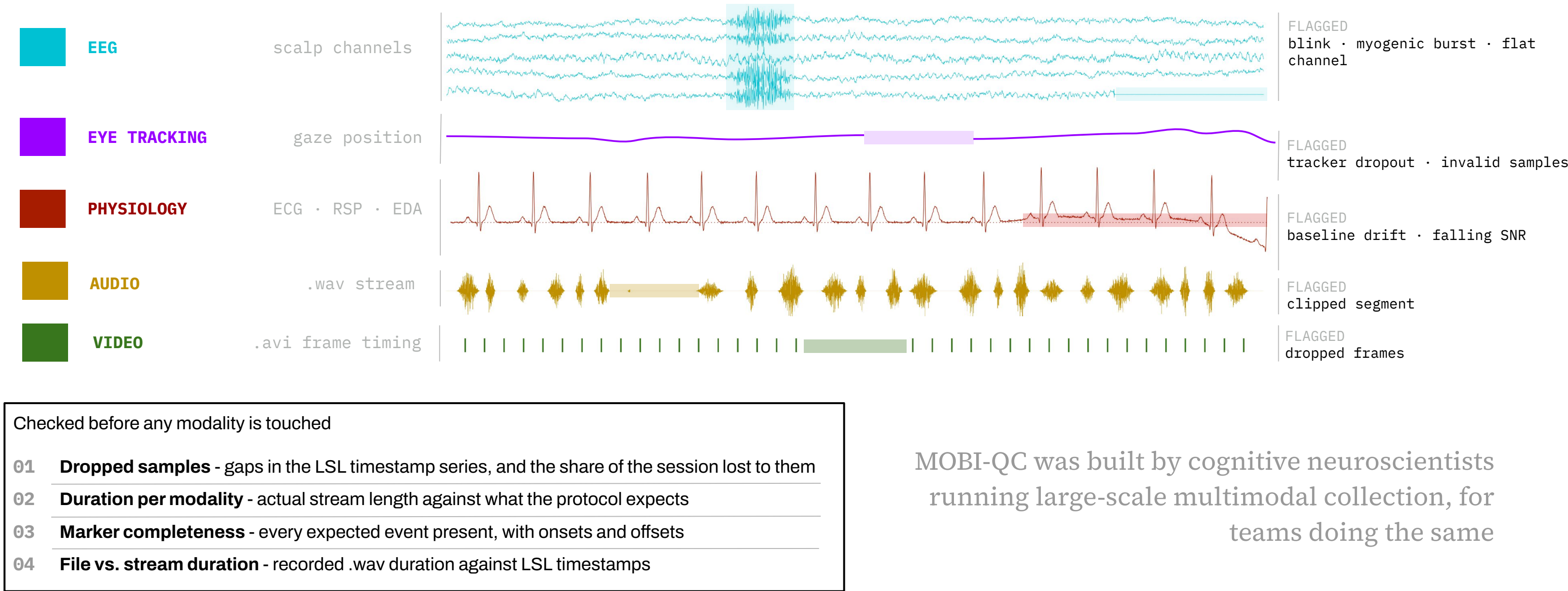
In practice, many teams begin large-scale collection without continuous safeguards. A pilot dataset that looks acceptable is what licenses full deployment; then subtle equipment malfunctions, sensor drift, and inter-experimenter variability introduce systematic artifacts over time.

Collected at scale and under time pressure, these issues go unnoticed until post hoc analysis - at which point remediation is costly or impossible. The losses are time, financial resources, and scientific validity.



One session, five streams, five silent failures

Some failures can be invisible at the acquisition console and unmistakable in the report.

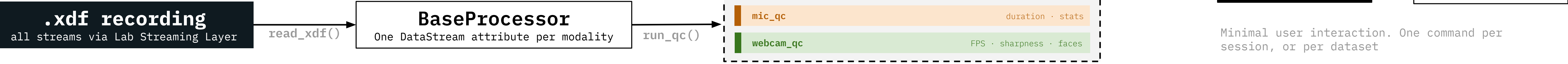


Methods

THE PIPELINE

Raw LSL recording in, quality report out

MOBI-QC ingests recordings acquired through the Lab Streaming Layer and runs a fully automated pipeline: streams are routed to modality-specific modules, each tailored to that signal’s properties and known failure modes, then aggregated into one report.



WHAT GETS CHECKED

A dedicated pipeline per modality

EEG `eeg_qc`

- Ocular and myogenic artifact annotated automatically
- Robust channel assessment — RANSAC via pyPREP; bad and interpolated channels
- ICA for data-driven noise removal; % dropped samples and % good
- 60 Hz noise removal before montage and robust reference

```
preprocess_eeg() · run_ICA() · annotate_blinks() · annotate_muscle()
```

Eye tracking `eyetracking_qc`

- Valid samples per eye; coordinate-system consistency
- 3D gaze validity; left/right validity difference
- Gaze-point differences over 0.02 mm; pupil diameter
- Euclidean L/R gaze distance, plotted across the session

```
validate_et_data() · compare_LR_validity() · check_xyz_validity() · check_coordinate_system_validity()
```

Peripheral physiology `ecg · respiration · eda`

- ECG — SNR, QRS peaks, spectral centroid, Welch baseline power
- Respiration — rate, breath amplitude, peak-to-peak interval, autocorrelation
- EDA — SCL / SCR features, response amplitude, coefficient of variation
- Signal-to-noise from residual noise power; baseline drift per stream

```
preprocess_ecg() · calculate_ecg_snr() · get_resp_peak_to_peak() · calculate_scr_features()
```

Audio `mic_qc · .wav`

- LSL-vs-file duration difference; amplitude histogram statistics
- Flags stray listening and social-task audio
- Quartile, mean, standard-deviation and range statistics across the recording

```
compare_lsl_wav_durations() · calculate_mic_stats()
```

Video `webcam_qc · .avi`

- Frame-rate stability; brightness; sharpness (Laplacian variance)
- % dropped frames; face-detection rate
- Mean brightness per frame; frames missing against nominal rate

```
calculate_FRS() · compute_sharpness() · detect_faces() · calculate_percent_dropped()
```

Task and Markers `behavior_qc`

- Every expected event marker present; durations vs. protocol
- Pre-stimulus interval; mean response time
- Stimulus onsets and offsets confirmed against the audio stream

```
get_missing_markers() · unexpected_durations() · confirm_prestim_interval() · calculate_mean_rt()
```

BEYOND ONE SUBJECT

Every session, read against the whole sample

MOBI-QC supports batch processing and produces group-level summaries describing the overall health of a dataset. Subject-level metrics are contextualized within the full sample, so outliers and emerging trends across participants, sessions, or collection sites are visible immediately.



Defaults you can run. Every metric was selected through an extensive review of the peer-reviewed literature, chosen to stay sensitive to meaningful degradation while remaining interpretable by applied research teams.

Thresholds are exposed for teams that want to be more or less conservative, but the whole pipeline runs end to end on validated defaults — standardized QC without expert intervention.

References:

[1] F. M. P. Vieira and J. P. S. Cunha, "Synchronization of Multimodal Physiological Data Streams: State-of-the-Art, Trends, and Future Challenges," in IEEE Access, vol. 14, pp. 70200-70223, 2026, doi: 10.1109/ACCESS.2026.3691002.

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